

Yixiang REN

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EDUCATION

Zhejiang University – University of Edinburgh Joint Institute (ZJU-UoE) — B.Sc. in Bioinformatics Hangzhou, China — 2023.09 – 2027.06 (expected)

Major GPA: 3.96/4.3 Ranking: 11/54 English: CET4: 611 & CET6: 617

Program Type: Two-degree (4+0)

RESEARCH EXPERIENCE

Project: Stromal Cell Roles in Pubertal Mammary Development Under High-Fat Diet

Undergraduate Researcher, Associated Professor: Dr. Chaochen Wang, Zhejiang University

2024.07 – Present

- Built an end-to-end snRNA-seq analysis workflow (QC - integration/batch correction - annotation) and quantified lineage dynamics of adipocyte subclusters under HFD vs control using trajectory/velocity frameworks (Monocle3/scVelo).
- Identified an ECM-remodeling stromal population with a distinct marker Adam12 consistent with recent literature published by Zemin Zhang and under functional validation.
- Assisted with IHC, immunofluorescence, mouse dissection, milk extraction.
- Revealed HFD-disrupted macrophage–epithelial signaling (Cholesterol-LIPA → Rora), providing mechanistic hypotheses for impaired ductal development.
- Accepted by ISSCR/Society of Developmental Biology 2026

Project 2: Human–Mouse Single-Cell Transcriptomic Database & Modeling

Undergraduate Researcher, Associated Professor: Dr. Chaochen Wang, Zhejiang University

2024.12 – Present

- Curated and processed 50+ public FASTQ-based scRNA-seq datasets (>1M cells) with standardized preprocessing (cellranger/Scanpy) and integration (scVI), enabling consistent cross-study comparisons.
- Implemented automated/assisted cell-type annotation and downstream analyses (DE, enrichment), and contributed to computational modeling for cross-species mapping (ortholog handling / harmonized labels).
- Contributed to web platform development (data organization, metadata schema, query/visualization modules); manuscript in preparation (analysis + platform module completed, target journal: Nucleic Acids Research).
- Web Page: <https://skin-scsaid.com/>

Project 3: Transformer powered TF Binding Prediction & Large-Scale Brain Cell Atlas & Sparsity-Aware Metacell Pipeline for Single-Cell Brain Multiomics Foundation Modeling

Research Assistant, PI: Dr. Xushen Xiong, Zhejiang University, Liang Zhu laboratory

2025.8 – Present

- Collected 400+ CTCF-related ChIP-seq datasets, assisted with Hi-C matrix prediction model (Hi-READ) which is under Transformer framework optimization & downstream model performance evaluation, Accepted by Cold Spring harbor Asia AI&Biology.
- Participate to build a 3.8M-cell brain atlas and performed cell–cell communication analysis including intracellular context factorization, gene expression module (GEMs) construction, perform flow analysis (flowsig) and perturbation analysis (perpty).
- Designing a novel multiomics foundation model pipeline for single-cell brain data from scratch, proposing a learnable VQ-VAE metacell aggregation layer that addresses data sparsity end-to-end with downstream transformer fine-tuning — a capability absent from existing tools.

ACTIVITIES

Volunteer Teaching – Rural Education Support Program

Dali Prefecture, Yunnan, China

2024.07.15 – 2024.07.24

Taught 4th–5th grade local children (most left-behind children). Delivered academic lessons and child-development activities; contributed to emotional support and local engagement.

AWARDS & HONOURS

Zhejiang University Scholarship, ZJU — 2024 (Rank 8/54)

Zhejiang University Scholarship, ZJU — 2025 (Rank 10/54)

TECHNICAL SKILLS

Programming & Tools

- Python, R, Linux, Shell scripting
- Scanpy, Seurat
- SPSS
- Adobe illustrator

Bioinformatics Expertise

- scRNA-seq, snRNA-seq
- Epigenomics (ChIP-seq, ATAC-seq, Hi-C)
- Spatial transcriptomics
- Regulatory network inference (pySCENIC)
- Cell–cell communication analysis
- Computational modeling

Wet-Lab Skills

- Immunohistochemistry (IHC)
- Immunofluorescence (IF)
- Mouse dissection
- Mouse milk extraction
- Western Blot
- PCR